

Total No. of Questions : 4]

SEAT No. :

PD-301

[Total No. of Pages : 1

[6411]-80

B.E. (Information Technology) (Insem.)

DISTRIBUTED SYSTEMS

(2019 Pattern) (Semester - VIII) (414450)

Time : 1 Hour]

[Max. Marks : 30

Instructions to the candidates:

- 1) *Answer Q1 or Q2, Q3 or Q4.*
- 2) *Figures to the right indicate full marks.*
- 3) *Neat diagrams must be drawn wherever necessary.*
- 4) *Assume suitable data, if necessary.*

- Q1)** a) Discuss the challenges faced by distributed systems due to heterogeneity and openness. **[7]**
- b) Resources in the World Wide Web and other services are named by URLs. What do the initials URL denote? Give examples of three different sorts of web resources that can be named by URLs. **[8]**

OR

- Q2)** a) Discuss the challenges presented while designing scalable distributed systems in detail. **[7]**
- b) Explain the Internet-scale and Contemporary generations of distributed systems with respect to scale, heterogeneity, and properties such as openness and quality of service. **[8]**
- Q3)** a) What is Remote Method Invocation? Explain remote interfaces with reference to JAVA RMI and CORBA system. **[7]**
- b) What is middleware in Distributed System? Explain intelligent and content centric middleware in brief. **[8]**

OR

- Q4)** a) What is Middleware? Differentiate between Architecture and Middleware. **[7]**
- b) Explain XML-based Web services middleware along with SOAP and REST protocols. **[8]**

